



**JAIDEV EDUCATION SOCIETY'S**  
**JD COLLEGE OF ENGINEERING AND MANAGEMENT**  
**KATOL ROAD, NAGPUR**

Website: [www.jdcoem.ac.in](http://www.jdcoem.ac.in) E-mail: [info@jdcoem.ac.in](mailto:info@jdcoem.ac.in)

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science and Engineering

*"A place to learn; A chance to grow"*

**2025-2026(Even Semester)**

**MSE-1 EXAMINATION**

**Course: B. Tech in CSE**

**Year/Semester: 6<sup>th</sup> Semester (3<sup>rd</sup> Year)**

**Subject: LLM Fine Tuning & Optimization**

**Subject Code:CS6M005**

**Max Marks: 20**

**Date: 10/02/2026**

**Duration: - 1 Hr.**

**Instructions to the Students:**

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

**Students should be able to:**

- CO 1:** Understand the architecture and working principles of Large Language Models.
- CO 2:** Apply different fine-tuning methods to adapt pre-trained LLMs for specific tasks.
- CO 3:** Implement parameter-efficient techniques like Adapters and LoRA to optimize resource use.
- CO 4:** Utilize optimization methods such as quantization and distillation for efficient deployment.
- CO 5:** Use modern frameworks and tools to train, fine-tune, and evaluate LLMs effectively.

**SECTION A**

1. All Questions are compulsory.
2. Answer in one word/sentence.

| Q. No. | Questions                                                        | Level/CO Mapped | Marks |
|--------|------------------------------------------------------------------|-----------------|-------|
| Q.1    | Name the two main components of an Encoder–Decoder architecture. | L1/CO1          | 1     |
| Q.2    | State pre-training in Large Language Models?                     | L1/CO1          | 1     |
| Q.3    | Define transfer learning in the context of LLM fine-tuning.      | L1/CO2          | 1     |
| Q.4    | Define Foundation Model?                                         | L1/CO1          | 1     |

**SECTION B**

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence).

|     |                                                                        |        |   |
|-----|------------------------------------------------------------------------|--------|---|
| Q.5 | Explain the role of the encoder in Large Language Models.              | L2/CO1 | 2 |
| Q.6 | Describe the decoder architecture and its function in text generation. | L2/CO1 | 2 |
| Q.7 | Differentiate between pre-training and fine-tuning.                    | L2/CO1 | 2 |
| Q.8 | Explain any one data-related challenge faced during LLM training.      | L2/CO2 | 2 |

**SECTION C**

1. Solve any two questions.

|      |                                                                                                                                                       |        |   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|
| Q.9  | Compare Encoder-only, Decoder-only, and Encoder–Decoder architectures used in LLMs with suitable examples.                                            | L4/CO1 | 4 |
| Q.10 | Explain Full Fine-Tuning vs. Feature-based Fine-Tuning in details with their advantages, limitations, and use cases.                                  | L4/CO2 | 4 |
| Q.11 | Describe the dataset preparation and pre-processing steps required for fine-tuning an LLM. Explain tokenization, cleaning, and validation techniques. | L4/CO2 | 4 |